Securities Lending and Information Acquisition*

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Abstract

We show that mutual fund managers exploit the securities lending market as an information acquisition channel. Using regulatory data from the Deutsche Bundesbank that records security-level lending positions for the universe of German open-end equity mutual funds, we find that active funds reduce their holdings in stocks they lend by 0.33 percentage points per month. The effect is absent for passive funds. To identify the information acquisition mechanism, we interact lending with mandatory short position disclosure events under EU regulation. The trading response to lending nearly triples when a disclosure event follows in the subsequent month, with interaction coefficients of −0.80 to −0.86 percentage points. Only the lead disclosure amplifies the effect; contemporaneous and lagged disclosure interactions are small and statistically insignificant. Information flows most strongly through same-manager fund lending within families and is amplified in concentrated lending markets with fewer active agents. Our findings show that the securities lending market serves as a conduit for private information about short-selling demand, enabling active fund managers to make more informed portfolio decisions.

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1 Introduction

When a mutual fund lends shares from its portfolio to a short seller, does the fund learn something valuable? Short sellers are among the best-informed participants in financial markets, and a large body of evidence shows that their trades predict negative future returns.[1] By participating in the securities lending market, fund managers observe, in real time and at the security level, which of their holdings are being borrowed. This information is private: aggregate short interest is published only with a delay, and position-level lending data are not publicly available. We ask whether active fund managers exploit this private information channel by adjusting their portfolios in response to lending activity, and, if so, how to identify the mechanism through which information is acquired and acted upon.

We study these questions using a unique regulatory dataset from the Deutsche Bundesbank that records the security-level lending positions of all German open-end equity mutual funds. Our sample covers 320 funds from 36 fund families over December 2014 to December 2018. Crucially, this period coincides with the European Union's mandatory short position disclosure regime, which requires public disclosure of net short positions exceeding 0.5% of a company's issued share capital (Jank and Smajlbegovic, 2015, Jank et al., 2020). This regime provides an institutional source of identification that has no direct analog in existing U.S. studies: we can test whether the private information embedded in lending activity predicts fund trading specifically before a large short position is publicly revealed, establishing a clean timing test of the information acquisition mechanism.

Our central contribution is to identify the information channel through which lending activity affects portfolio decisions. The key source of identification is the interaction between a fund's private lending signal and the subsequent mandatory disclosure of a large short position. If active lenders are extracting information from the pattern of borrowing demand, the position reduction they undertake should be amplified precisely when that demand precedes

[1] See, among others, Diamond and Verrecchia [1987], Boehmer et al. [2008], Engelberg et al. [2012], and Karpoff and Lou [2010].

a public confirmation event—a new disclosure that the market can observe. We find exactly this pattern.

Our main empirical findings are as follows.

First, active funds reduce their positions in stocks they lend. A one-standard-deviation increase in lending activity is associated with a 0.33 percentage point reduction in the monthly position growth rate, and this effect is absent for passive funds, who cannot freely adjust holdings (Table 4). The active-passive split is the primary placebo test: passive funds hold the same stocks and participate in the same lending market, but their portfolios are mechanically constrained. The fact that only active funds respond rules out mechanical explanations such as share recalls for proxy voting or collateral-driven rebalancing, which would affect both fund types equally.

Second, we confirm that lending conveys information using mandatory disclosure events as an external validation. Within the disclosure sample—stocks that eventually receive a public large short-position disclosure—the baseline position reduction for active lenders is -0.28 percentage points per month (Table 5, column 1). This effect grows to -0.80 to -0.86 percentage points when the lending activity in month $t-1$ precedes a new disclosure event in month $t+1$, nearly tripling the magnitude. Critically, the amplification is *asymmetric in time*: only the lead disclosure indicator amplifies the effect; contemporaneous and lagged disclosure interactions are small and statistically insignificant (Table 7). Passive funds show no response at any horizon (Table 5, column 8). This temporal pattern—pre-announcement reductions by lenders, no post-announcement adjustment—is inconsistent with a mechanical story and is consistent only with active information acquisition through the lending process.

Third, information flows are strongest within fund families, with the same portfolio manager's other lending funds generating the largest incremental reduction (Table 8). The effect attenuates with organizational distance: lending by other managers within the same family generates a smaller but still detectable response, consistent with centralized lending desks serving as conduits for within-family information sharing (Pool et al., 2012). We use passive fund

lending in the same family as an alternative, plausibly more exogenous measure of lending demand (Table 6), and the results are robust; a formal IV design with a reported first stage is deferred to the revision.

Fourth, the informativeness of the lending signal is amplified in concentrated markets (Table 9). When fewer agents intermediate lending for a given stock—measured by the number of active lending agents, the family's HHI of lending concentration, or the inverse of the number of stocks lent—each lending contract carries more signal about the identity and size of the short position being built. These heterogeneity results sharpen the economic interpretation: information acquisition is not uniform but varies predictably with market structure.

These findings speak to a central tension in the securities lending literature. Evans et al. [2017] (hereafter EFP) document that active funds that participate in lending underperform otherwise similar non-lending funds by 0.5%–0.7% per year, a puzzle they attribute to fund family incentives that prevent managers from acting on short-selling demand signals. Our results are consistent with managers responding to the lending signal when they have portfolio discretion, and the EU disclosure design allows us to confirm that the mechanism is information rather than mechanics. Whether this trading response is sufficient to overcome the overall cost of lending participation—and thereby resolve the EFP underperformance puzzle—requires a direct performance comparison that we defer to the revision; the underperformance documented by EFP may reflect institutional constraints, lending revenue drag, or both.

Our paper is most closely related to Honkanen [2025], who uses hand-collected U.S. SEC filings on the ten largest fund families to study securities lending and trading at the quarterly frequency. Honkanen [2025] establishes that active funds reduce portfolio weights in lent stocks and that information spills over within fund families from passive lenders to active traders. We extend this work in three ways that jointly constitute our main contribution. First, our Bundesbank data cover the *universe* of German funds rather than the ten largest U.S. families, eliminating sample-selection concerns. Second, we observe lending at the monthly rather than quarterly frequency. Third—and most importantly—the EU mandatory short position disclo-

sure regime provides an independent, institutional identification of the information channel. The disclosure interaction test cannot be conducted in the U.S. setting because U.S. short position data are not reported at the position level with the same detail and timeliness. Without an external validation of this kind, it is difficult to rule out alternative explanations for the lending-trading link; our paper provides it.

Nurisso [2026] provides the theoretical foundation for our findings. In his model, active lenders acquire private information by observing that a short seller is willing to borrow shares. This creates a fundamental commitment problem: learning generates an incentive to recall the shares and sell, which short sellers anticipate. Active lenders must therefore raise lending fees to credibly commit against recall, making them less attractive counterparties. Passive lenders—unable to discretionarily trade—face no such constraint, offer lower fees, and become the preferred counterparty for informed short sellers (Palia and Sokolinski, 2024). In concentrated lending markets, fewer intermediaries observe borrowing demand per transaction, making each lender's signal more precise. Our results map directly onto all three predictions: active funds reduce positions in lent stocks while passive funds do not; the response is amplified when lending precedes a mandatory public disclosure; and the effect is strongest in concentrated lending markets. Chague et al. [2025] identify a distinct but related channel in the Brazilian market: brokers intermediating securities lending transactions leak information about skilled short sellers' directional bets to their other clients. Our paper demonstrates that the lender *itself* acquires and acts on the information, even without broker-mediated leakage.

We also contribute to the literature on short-sale constraints and price efficiency (Saffi and Sigurdsson, 2011, Porras Prado et al., 2016), information production in financial markets (Goldstein and Yang, 2017, Grossman and Stiglitz, 1980), and short position disclosure (Jank et al., 2020, Jones et al., 2016). Prado [2015] shows that institutions buy shares in response to increases in lending fees; our paper documents the complementary short-side signal: when borrowing demand is high, active managers reduce their exposure to avoid capital losses. Cohen et al. [2007] establish that demand-side shifts in the borrowing market—rather than supply-

side constraints—predict negative returns, directly motivating why the private demand signal observed through lending conveys valuable information.

The organization of the paper is as follows. Section 2 describes the institutional background and data. Section 3 presents the empirical strategy. Section 4 reports the baseline results on position changes by active and passive funds. Section 5 identifies the information acquisition channel using the disclosure regime. Section 6 examines within-family information flows. Section 7 analyzes market structure heterogeneity. Section 8 concludes.

2 Institutional Background and Data

2.1 German Securities Lending Regulation and Comparison with the United States

The German securities lending framework and the U.S. framework share the same statutory core. In the United States, securities lending by registered investment companies is governed by guidance under the Investment Company Act of 1940 and informal SEC guidance (Division of Investment Management Guidance Update No. 2014-06), requiring full collateralization of loans, conservative collateral reinvestment, and preserved recall rights; U.S. funds may lend at most one-third of total assets [Honkanen, 2025]. In Germany, Sections 200 to 202 of the *Kapitalanlagegesetzbuch* (KAGB) impose closely analogous requirements: (i) lending transactions with any single counterparty must not exceed 10% of a fund's net asset value; (ii) funds must receive collateral worth at least 100% of the loan value; (iii) collateral must be held in high-quality sovereign debt, short-term money market funds, or repurchase agreements; (iv) remuneration must reflect current market conditions; and (v) funds must be able to terminate any loan at any time. Both systems therefore guarantee that lending is fully collateralized and callable on demand, making loan recall feasible regardless of short-selling pressure.

The two markets differ in two economically relevant respects. First, European UCITS rules require that 100% of net lending revenue — after operational costs — be returned to the lending

fund, while U.S. fund boards have discretion over the revenue split between the fund and its lending agent. In practice, German and European funds receive between approximately 50% and 95% of gross revenue [Voicu and Prache, 2019], while U.S. splits typically run 60%–80% fund / 20%–40% agent [Evans et al., 2017]. Second, the dominant collateral type differs: cash collateral accounts for roughly 80% of U.S. securities loans, while non-cash collateral (primarily sovereign debt) accounts for approximately 60% of European loans [Jank and Smajlbegovic, 2015, Financial Stability Board]. These differences affect the economics of lending but not its information content.

The most consequential institutional difference for the information channel we study is the lending-agent structure. In the United States, most securities lending is intermediated by third-party custodian banks — subsidiaries of State Street, BNY Mellon, and JPMorgan — that manage lending programs for multiple unrelated fund families and do not disclose borrower identity to the fund manager [Baklanova et al., 2016]. In Germany, by contrast, larger fund families typically operate in-house lending desks — organizational units within the fund management company (*Kapitalverwaltungsgesellschaft*, KVG) — that pool lending across funds in the family and negotiate directly with prime broker counterparties. This in-house structure creates organizational proximity between the entity observing borrowing demand and the portfolio managers who make trading decisions. The identity of the short-seller counterparty, the size of the loan, and the terms of the borrowing arrangement are visible within the same organizational entity that manages the portfolio. Honkanen [2025] documents that U.S. active mutual funds also reduce positions in lent stocks, suggesting that even third-party intermediation does not fully eliminate information transmission; but the transmission channel is more direct in the German in-house model, which is consistent with our finding that the information effect is strongest through same-manager lending.

The most important regulatory difference for identification is the short-position disclosure regime. In the United States, FINRA publishes biweekly aggregate short interest reports by security, with no identification of individual short sellers regardless of position size, and with

a publication lag of approximately one to two weeks. In the European Union, Regulation (EU) No. 236/2012 requires any investor whose net short position exceeds 0.5% of a company's issued share capital to publicly disclose the position — including the investor's identity, position size, and relevant issuer — to the national competent authority (BaFin in Germany) by the next business day. This creates a uniquely granular, near-real-time record of large short-seller positions with no direct U.S. analogue. Positions above 0.2% trigger a private notification to BaFin but are not publicly disclosed; positions above 0.5% are posted publicly on the BaFin website. Jank et al. [2020] document that short sellers strategically pile up positions just below the 0.2% private notification threshold, with an estimated excess mass of 92.8% below that threshold for first-time disclosers, consistent with informed short sellers placing significant value on maintaining the secrecy of their positions before disclosure.

This disclosure architecture is the foundation of our identification strategy. Before a short position crosses the 0.5% public disclosure threshold, the private information that a large short seller is accumulating a position exists only in the lending book of the fund lending the shares. The moment that position becomes public, the information transitions from private to common knowledge: Jones et al. [2016] show that public disclosures under Regulation 236/2012 reduce short interest and bid-ask spreads, confirming that the disclosure event is informative. The EU regime thus creates a precisely datable stock-level information event — the transition from private to public knowledge — that allows us to test whether active fund managers who observe borrowing demand through lending trade in anticipation of the disclosure. Because U.S. short interest reporting is aggregate, anonymous, and biweekly, no equivalent scheduled information transition exists in the U.S. context. The German regulatory environment is therefore uniquely suited to our identification strategy, while the core mechanism — funds acting on private information from lending activity — is broad enough to generalize, as documented by Honkanen [2025] in U.S. data and consistent with the strategic borrowing equilibrium studied by Palia and Sokolinski [2024], in which short sellers prefer lending from passive investors precisely because such funds do not extract and act on the lending signal.

2.2 Data Sources

Our data on German mutual funds comes from the Investment Funds Statistics (IFS) of the Deutsche Bundesbank, covering December 2014 through December 2018. Since September 2009, all investment companies domiciled in Germany must report general fund information and portfolio holdings to the IFS. The data are collected monthly and provide detailed end-of-month, security-by-security information about each fund's portfolio. For each holding, a fund reports the ISIN, the number of shares held, and the nominal and market value of the position.[2]

In December 2014, the Bundesbank introduced a major reporting change requiring funds to disclose securities lending activities. For each security in the portfolio, each fund must report the end-of-month nominal value of any position held in a lending transaction.[3] This regulatory change provides us with a unique dataset: security-level lending positions for the universe of German investment funds.

We merge the IFS data with Morningstar fund information and aggregate the data to the fund level, matching share class ISINs to the fund identifier provided by Morningstar. For funds with multiple share classes, we calculate fund returns and expenses as asset-weighted averages across share classes. For assets under management, fund flows, and fund holdings, we aggregate across all share classes. Fund age is based on the inception date of the oldest share class.

We combine the IFS mutual fund data with stock information from Datastream, securities lending data from IHS Markit, and the Fama-French risk factors from Kenneth French's data library.[4]

[2] Dötsch et al. [2018] provide a detailed description of the IFS data.

[3] Securities lending information is reported at the fund-security level and is therefore constant across all share classes of an individual fund.

[4] Following Ferreira et al. [2018], we estimate four-factor alphas using regional factors based on a fund's investment region and use world factors for global funds.

2.3 Sample Construction

The initial IFS-Morningstar matched sample consists of 2,044 funds from 58 fund families. We apply several filters. First, we exclude closed-end funds and alternative investment funds. Second, we remove non-equity, sector, and emerging market funds based on their Morningstar category. Third, we keep only funds older than three years with more than $10 million in assets under management to mitigate potential incubation biases [Evans, 2010]. Our final sample consists of 320 German open-end equity mutual funds from 36 fund families.

We further restrict our sample to holdings of securities identified as common or preferred equity, excluding approximately 2% of observations. We retain only securities traded in the 23 developed markets listed in Fama and French [2012], excluding another 2% of observations. We drop observations where the reporting date of the holding in the IFS data exceeds the stock's delisting date in Datastream (less than 0.5% of observations) and remove fund-month pairs where a fund closes all positions, as we assume the fund is liquidated (less than 0.5% of observations).

2.4 Variable Construction

We use the Bundesbank holdings data to construct two variables capturing the monthly trading activity of a fund. $PosChange_{i,j,t}$ is the percentage change in the number of shares of stock i held by fund j in month t, adjusted for stock splits and winsorized at the 99.5% level. For securities lending activity, we compute $sh_sec_lend_{i,j,t}$, defined as the fraction of a particular portfolio position in stock i lent by fund j in month t.

At the stock level, we measure aggregate shorting demand as the value of securities on loan scaled by the total market capitalization of that security ($OnLoan$), obtained from IHS Markit.

2.5 Descriptive Statistics

Table 1 presents the breakdown of our sample by investment focus. Of the 320 funds, 267 are actively managed and 53 are passively managed. In terms of investment focus, 92 funds have a global focus, 135 focus on European equities, and 58 focus on German equity, with total assets under management of EUR 156.75 billion.

Table 2 provides summary statistics on the securities lending behavior of German investment funds. A third of all funds participate in securities lending during our sample period. The share of securities lenders is considerably lower for actively managed funds (25.47%) compared to passive funds (69.81%). Investment funds typically lend only a small fraction of their portfolio, with an average of 3.8% of AUM on loan. These figures are comparable to U.S. mutual fund lending practices documented in Porras Prado et al. [2016] and the Investment Company Institute [Grohowski and Collins, 2014].

Table 3 presents descriptive statistics for the monthly fund-security panel data. Panel A focuses on portfolio characteristics. Portfolios are well-diversified, with a mean (median) portfolio holding of 1.08% (0.63%). Approximately 2% of fund-security positions are lent in a given month ($D(\text{Lender})_{i,j,t}$). Conditional on lending, the mean position lent represents 6.86% of the stock's total market capitalization ($\text{OnLoan}_{i,j,t}$). At the fund-family level, 5% of fund-security positions have at least one family member lending the stock, with a conditional mean of 7.65% of market capitalization ($\text{OnLoan}_{i,f,t}$).

Panel B presents stock-level characteristics. For stocks held by German investment funds, total shorting demand (*OnLoan*) has a mean of 2.46% and a median of 1.05% of market capitalization. Average lending supply is 18.18% and the mean short fee is 65.54 basis points per annum, reflecting the presence of some high-fee "specials." Mandatory public short-position disclosures under EU Regulation 236/2012 are active for 19% of stock-month observations (D(Disclosed Short Position)), while new disclosure events — the first public crossing of the 0.5% threshold — occur in 0.4% of stock-months (D(Disclosure Event)).

Panels C and D report fund-level and fund family-level characteristics, respectively. In our sample, 85% of observations come from active funds and 15% from passive funds. Overall, 20% of observations are from funds that participated in the securities lending market in a given month. At the family level, 17% of observations come from families with at least one fund participating in lending.

3 Empirical Strategy

3.1 Empirical strategy

We examine how short selling demand signals in the equity lending market affect mutual fund portfolio allocation. Consider the following general characterization of mutual fund holdings, where $Y_{i,j,t}$ is the holding for fund j in stock i at time t:

$$Y_{i,j,t} = f\left(Y_{i,j},\ Y_{i,t},\ Y_{j,t},\ \phi_{i,j,t}\right) \tag{1}$$

Mutual fund holdings are an equilibrium outcome determined by four components: a fund-stock time-invariant component $Y_{i,j}$, which reflects a fund's persistent preferences for a stock based on investment philosophy or style; a common stock-time component $Y_{i,t}$, which captures signals common to all investors including aggregate short selling demand; a fund-time component $Y_{j,t}$, which captures fund-level investment style, portfolio breadth, and governance; and a fund-stock-time specific component $\phi_{i,j,t}$.

Our focus is on the fund-stock-time component $\phi_{i,j,t}$: how fund j's specific demand for stock i changes when the fund receives a *private* short selling demand signal $OnLoan_{i,j,t-1}$ through the equity lending market. For example, consider a fund active in the lending market that lends $OnLoan_{i,j,t-1}$ shares of stock i at time $t-1$. The amount on loan reveals information

about short selling demand that is not contained in public signals—because public signals are absorbed by $Y_{i,t}$—but is observed privately by the fund through its lending contracts.[5]

In a saturated fixed effect estimation, we estimate changes in fund-stock portfolio holdings as a function of a fund's lending activity, including fund-stock, stock-time, and fund-time fixed effects ($\alpha_{i,j}$, $\alpha_{i,t}$, and $\alpha_{j,t}$). These fixed effects absorb public stock-time price signals common to all investors, fund-time information and investment decisions common to all stocks in a fund's portfolio, and fund-stock time-invariant preferences.

To identify the effect of the private signal $OnLoan_{i,j,t-1}$ on portfolio allocation, we estimate:

$$Y_{i,j,t} = \alpha_{i,j} + \alpha_{i,t} + \alpha_{j,t} + \beta_1 \, OnLoan_{i,j,t-1} + \mathbf{X}'_{i,j,t-1} \, \gamma_1 + \epsilon_{i,j,t}, \tag{2}$$

where $Y_{i,j,t}$ is the monthly percentage change in the number of shares of stock i held by fund j at time t ($PosChange_{i,j,t}$). $OnLoan_{i,j,t-1}$ is the dollar value of shares lent by fund j in stock i at $t-1$ divided by the market capitalization of stock i, standardized to mean zero and unit standard deviation to facilitate comparison of effect magnitudes across specifications.[6] This measures the fund's exposure to private short selling demand. $\mathbf{X}_{i,j,t-1}$ includes lagged position weight ($PosWeight_{i,j,t-1}$) and an Amihud [2002] illiquidity rank ($ILLIQ_{i,j,t-1}$). Standard errors are clustered at the fund level.[7] Variable definitions are provided in Appendix A.

The coefficient β_1 identifies the effect of the private signal: we predict $\beta_1 < 0$ if lenders act on the information embedded in short selling demand. The counterfactual is funds that hold the same stock but do not participate in lending and therefore receive no private signal. Any effect of aggregate short selling demand $OnLoan_{i,t}$ is absorbed by $\alpha_{i,t}$.

[5] A measure of aggregate short selling demand is available to institutional investors. Brokers often provide data on special lists and borrowing costs to institutional investors with substantial trading volume. Markit reciprocally shares lending market information with participating clients.

[6] Standardization is performed within the estimation sample. All *OnLoan* variables throughout the paper are standardized in this way unless stated otherwise.

[7] For the baseline specification in equation (2), clustering at the fund level is appropriate because the primary source of residual dependence is within-fund correlation across the security and time dimensions; the identifying variation is fund-specific rather than time-specific. For specifications involving the mandatory disclosure events in Sections 5–7, we double-cluster by fund and time. Disclosure events are correlated across funds in the same month, introducing a time-series dimension to the error structure that a single fund cluster does not fully capture.

3.2 Active vs. passive funds

A critical identification test is whether the trading response to lending signals is confined to active fund managers who can freely adjust portfolios, rather than passive fund managers who must track a benchmark index. We estimate:

$$Y_{i,j,t} = \alpha_{i,j} + \alpha_{i,t} + \alpha_{j,t} + \beta_1 \, OnLoan_{i,j,t-1} + \beta_2 \, OnLoan_{i,j,t-1} \times Active_j + \mathbf{X}'_{i,j,t-1} \, \gamma_1 + \epsilon_{i,j,t}, \tag{3}$$

where $Active_j$ equals one for actively managed funds. The coefficient β_1 captures the response for passive funds, which we expect to be zero since passive funds cannot trade on information. β_2 captures the incremental response for active funds relative to active non-lenders; our information acquisition hypothesis predicts $\beta_2 < 0$. Note that aggregate short selling demand $OnLoan_{i,t-1}$ is absorbed by the stock×time fixed effect $\alpha_{i,t}$, so the coefficients identify the fund-specific private signal over and above any aggregate demand common to all funds.

3.3 Identification via mandatory short position disclosures

Our primary identification strategy exploits mandatory public disclosures of large short positions. Under Article 9 of Regulation (EU) No 236/2012, any net short position exceeding 0.5% of issued share capital must be publicly disclosed the following trading day. Above 0.5%, further disclosures are required at 0.1% increments. The regulation applies to all investors irrespective of origin, with standardized disclosure across EU countries.[8]

We focus on *new* disclosure events—stocks that have not had any public short position disclosure in the preceding six months. This restriction isolates the most informative public signals: follow-on disclosures for stocks with known short positions are less informative [Jones et al., 2016]. These disclosed positions predict negative returns [Jank and Smajlbegovic, 2015]

[8]The regulation and relevant data are described in detail in Jank and Smajlbegovic [2015], Jones et al. [2016], and Jank et al. [2020]. Publicly disclosed short positions are more frequently observable for large and liquid stocks [Jank et al., 2020], consistent with our sample.

but do not fully reveal the private structure of shorting demand below the disclosure threshold [Jank et al., 2020].

Selection concerns. The main identification challenge is that lending activity may be endogenous to portfolio decisions. A fund might reduce a position precisely because it can no longer lend shares (liquidity-driven), or might lend more shares in positions it plans to reduce (selection). The three-way fixed effects structure directly addresses this concern: $\alpha_{i,t}$ absorbs stock-time common shocks including aggregate borrowing demand, $\alpha_{j,t}$ absorbs fund-level portfolio rebalancing, and $\alpha_{i,j}$ absorbs time-invariant fund-stock preferences. The identifying variation is the idiosyncratic deviation of a fund's lending in a specific stock at a specific time, relative to what is predicted by these three layers of fixed effects.

A related mechanical concern is that the fund×security fixed effect $\alpha_{i,j}$ demeans $OnLoan_{i,j,t-1}$ within each cell, and because lending is zero in most observations (only 2% of fund-security-month observations have any shares on loan), the demeaned residual is positive in lending months and negative otherwise. If a fund mechanically reduces positions as lending terminates—with no informational content—this autocorrelation structure could generate a spurious negative $\hat{\beta}_1$. Three features of the design collectively rule out this mechanical story. First, and most directly, passive funds are subject to the identical fund×security fixed effect structure and have substantially *higher* lending rates (69.8% of passive funds participate in lending versus 25.5% of active funds), yet they exhibit precisely zero trading response (Table 5, column 8). A mechanical channel would affect passive funds at least as strongly as active funds, since the demeaning algebra is identical and passive funds' lending is more prevalent. Second, the disclosure timing interaction is forward-looking: the mechanical channel predicts no amplification when lending in month $t-1$ precedes a *future* disclosure event in month $t+1$, because the mechanical story has no access to forward-looking public information. Yet the interaction $OnLoan_{i,j,t-1} \times Disclosure_{i,t+1}$ is the strongest and most significant term in Table 7. Third, the mechanical story predicts symmetric effects around any systematic shock to the stock—not the asymmetric lead-only pattern documented in Section 5. We plan to provide

formal diagnostics (excluding lending-termination observations; AR(1) residual tests) in the revision.

We further address endogeneity by exploiting the timing of mandatory disclosures. If the lending signal is informative, active lenders should reduce positions *before* a large short position is publicly disclosed—not after. We test this using the following regression:

$$\begin{aligned} Y_{i,j,t} = {} & \beta_1 \, OnLoan_{i,j,t-1} + \beta_2 \, Disclosure_{i,t+1} \times OnLoan_{i,j,t-1} \\ & + \beta_3 \, Disclosure_{i,t} \times OnLoan_{i,j,t-1} + \beta_4 \, Disclosure_{i,t-1} \times OnLoan_{i,j,t-1} \\ & + \mathbf{X}'_{i,j,t-1} \, \gamma + \alpha_{i,t} + \alpha_{j,t} + \alpha_{i,j} + \epsilon_{i,j,t}, \end{aligned} \tag{4}$$

where $Disclosure_{i,t}$ is an indicator equal to one if there is a *new* public short position disclosure in month t for stock i. $Disclosure_{i,t+1}$ and $Disclosure_{i,t-1}$ are the lead and lag of this indicator, respectively. The key coefficient is β_2: lenders reduce positions in the month *before* disclosure. We expect $\beta_3 = 0$ and $\beta_4 = 0$ because lenders' information advantage evaporates once positions are publicly disclosed.

This timing pattern—pre-announcement selling by lenders but not by non-lenders—is inconsistent with mechanical explanations (e.g., recall of lent shares to vote) and is most consistent with active information acquisition through the lending process. Standard errors are double-clustered at the fund and time level for all disclosure-sample specifications.

Contribution. Our approach builds on and extends the existing literature. Evans et al. [2017] document that securities lending generates revenues for mutual funds but do not examine whether lending conveys information. The theoretical framework in Nurisso [2026] provides our primary identification motivation: active lenders acquire private information but face a commitment problem, so that the information value of lending should be detectable precisely in the window before private information becomes public, and should be larger in concentrated markets where each lender's signal is more precise. We provide empirical confirmation of these theoretical predictions using three identification advances. First, we use European regulatory

data to directly link individual lending contracts to portfolio decisions, isolating the fund-level private signal. Second, we use mandatory disclosure events as the key external validation: if active lenders exploit private information, position reductions should be concentrated immediately *before* the information becomes public—the direct test of the Nurisso information channel. Third, by examining market structure—lending concentration, active agent counts, and portfolio breadth—we identify conditions under which the information value of the lending signal is highest, consistent with Nurisso's result that competition among lenders governs information precision. These market structure results are novel to the literature.

4 Securities Lending and Portfolio Decisions

We begin our analysis by examining whether mutual funds adjust their portfolio positions in response to securities lending activity. Our baseline specification, described formally in Section 3, regresses monthly position changes on lagged lending indicators while absorbing unobserved heterogeneity through a saturated set of fixed effects.

4.1 Baseline Results

Table 4 presents the baseline results. The dependent variable is $PosChange_{i,j,t}$, the monthly percentage change in the number of shares of stock i held by fund j at time t. Our key explanatory variable is $OnLoan_{i,j,t-1}$ (standardized), which measures the dollar value of shares lent by fund j in stock i as a fraction of market capitalization at $t-1$. All specifications include security×time, fund×time, and fund×security fixed effects. Standard errors are clustered at the fund level.

In column (1), we estimate the relationship for the full sample of funds. The coefficient on $OnLoan_{i,j,t-1}$ is -0.2069 and statistically significant at the 1% level ($t = -4.23$). Funds reduce their positions in stocks they lend. This effect is economically meaningful: a one-standard-

deviation increase in lending activity is associated with a 0.21 percentage point reduction in position size in the subsequent month.

4.2 Active vs. Passive Funds

A core prediction of the information acquisition hypothesis is that only active fund managers—who can freely adjust their portfolios—should respond to the lending signal. Passive fund managers, constrained to track their benchmark index, should not trade on the information even if they observe it.

Column (2) of Table 4 tests this prediction by interacting $OnLoan_{i,j,t-1}$ with an indicator for actively managed funds, as in equation (3). The interaction term $Active_j \times OnLoan_{i,j,t-1}$ is -0.3382 and highly significant ($t = -5.33$), while the uninteracted $OnLoan_{i,j,t-1}$ coefficient becomes small and statistically insignificant (0.0133, $t = 0.80$). The position reduction is entirely driven by active funds: passive funds do not adjust their portfolios in response to lending signals.

Columns (3)–(5) restrict the sample to active funds only. Column (3) confirms the baseline effect with a coefficient of -0.3271 ($t = -5.39$). Column (4) replaces the continuous lending measure with the extensive margin indicator $D(Lender)_{i,j,t-1}$, equal to one if the fund lent any shares of the stock in the prior month. The coefficient of -1.9165 ($t = -4.18$) indicates that active lending funds reduce positions significantly more than non-lending active funds. Column (5) includes both measures simultaneously; the continuous $OnLoan$ measure dominates (-0.3206, $t = -5.49$), while the extensive margin indicator becomes insignificant (-0.0584, $t = -0.15$), suggesting that the *intensity* of lending—not merely lending participation—drives the trading response.

Taken together, the results in Table 4 establish that active fund managers reduce positions in stocks they lend, with the effect concentrated at the intensive margin of lending. The passive fund placebo (column 2) rules out pure mechanical explanations such as share recall for voting purposes, which would affect passive and active funds similarly. This active-passive asymmetry

is the central empirical prediction of Nurisso [2026]: active lenders face a commitment problem that arises precisely from their ability to trade on the private signal, leading short sellers to prefer passive lenders as counterparties. The corollary for our tests is clean: passive funds, who cannot exploit the signal, exhibit zero response, while active funds respond strongly.

5 Information Acquisition Channel

The baseline results establish that active funds reduce positions in stocks they lend. We now investigate the mechanism: do funds trade on information acquired through the lending process? The model in Nurisso [2026] provides a precise prediction: lenders observe that a short seller is willing to borrow, which is informative about the stock's future value. This private signal is valuable precisely *before* the short seller's position crosses the mandatory public disclosure threshold. Once disclosed, the information is public and the lender's informational advantage evaporates. We exploit mandatory short position disclosure events—as described in Section 3.3—as an external validation of this mechanism. The key identification test is whether active lenders reduce positions specifically in the month *before* a large short position is publicly disclosed, consistent with exploiting private information embedded in the lending contract that is not yet visible to the market.

5.1 Mandatory Short Position Disclosure

Table 5 presents the results, estimated on the disclosure sample (observations within ± 3 months of a first-time large short position disclosure for stocks that are ever publicly disclosed). The dependent variable is $PosChange_{i,j,t}$. $OnLoan_{i,j,t-1}$ is standardized lending activity; $Disclosure_{i,t+1}$ equals one if stock i has a *new* public short position disclosure in month $t+1$ (defined as having no prior disclosure in the preceding six months). All specifications use active funds and double-cluster standard errors by fund and time.

Column (1) reports the baseline lending effect for this restricted sample: $OnLoan_{i,j,t-1} = -0.2837$, $t = -2.85$, confirming the baseline result holds in the disclosure sample. Column (2) adds the disclosure indicator, which is positive but insignificant (0.3633, $t = 1.25$), consistent with an absence of general reaction to impending disclosures among both lenders and non-lenders alike (this effect is absorbed by the stock×time fixed effects in the saturated specification).

Column (3) introduces the key interaction: $OnLoan_{i,j,t-1} \times Disclosure_{i,t+1}$. The interaction coefficient is -0.8027 and statistically significant at the 5% level ($t = -2.68$). Active funds reduce their positions substantially more—by an additional 0.80 percentage points per standard deviation of lending—when their lending activity precedes a new disclosure event by one month. Column (4) adds the full saturated three-way fixed effects; the interaction coefficient strengthens to -0.8622 ($t = -2.22$), allaying concerns that the result is driven by unobserved stock-time or fund-time factors.

Columns (5)–(7) replicate columns (1)–(3) using a lower-dimensional fixed effects structure (separate security, fund, and time fixed effects rather than their pairwise interactions); the results are qualitatively identical, confirming robustness to the fixed effects specification.

Column (8) provides the critical placebo test. Estimating the identical specification for passive funds only, the coefficient on $OnLoan_{i,j,t-1}$ is economically small and insignificant (0.0208, $t = 0.23$), and the interaction with $Disclosure_{i,t+1}$ is precisely zero (0.0258, $t = 0.28$). Passive funds show no trading reaction to lending signals around disclosure events. This placebo result strongly supports the information acquisition interpretation: only funds with discretion to trade on private information do so.

5.2 Reduced-Form Robustness: Passive Fund Lending

A potential concern is that a fund's own lending activity is endogenous to its trading decisions—a fund might selectively reduce positions in stocks it plans to stop lending, creating a spurious negative correlation. As a reduced-form robustness check, we replace $OnLoan_{i,j,t-1}$

with $OnLoan_{i,f\text{-passive},t-1}$, the aggregate lending of stock i by passive funds in the same family (Table 6). Since passive funds track indices and cannot adjust portfolio weights discretionarily, their lending is governed by portfolio composition and borrower demand rather than by active rebalancing decisions. This substitution provides a robustness check because the passive lender's decision to lend a given stock is plausibly unrelated to the active manager's concurrent trading plans.[9]

The coefficient on passive fund lending is -0.3406 ($t = -3.20$) in column (1), comparable to the baseline. The interaction with $Disclosure_{i,t+1}$ in column (3) is -1.1240 and significant at the 1% level ($t = -3.29$)—larger in magnitude than the baseline interaction (-0.8027). The larger magnitude is consistent with passive fund lending providing a signal that is less contaminated by the active manager's own rebalancing decisions, though part of the difference could also reflect measurement-error attenuation in the fund-level measure that is reduced by family aggregation. Column (4) confirms robustness with saturated fixed effects (-0.9156, $t = -2.32$). These results are consistent with the trading response being robust to this alternative lending measure, though they do not fully resolve the endogeneity concern in the absence of a formal IV structure.

5.3 Timing Tests: Leads and Lags of Disclosure

If active funds genuinely acquire information through lending and trade in anticipation of future short-selling pressure, only the *lead* disclosure should predict trading. Contemporaneous and lagged disclosure interactions should be small once the information has become public.

[9]This specification is a reduced-form regression, not a two-stage instrumental variables estimator. We do not report a first stage here because doing so requires a maintained exclusion restriction that passive fund lending affects active fund trading *only* through within-family information transmission—a restriction that is economically plausible but is not directly testable. Potential violations include correlated borrower demand (a short seller approaching family f simultaneously negotiates with both passive and active funds in the family) and common benchmark overweights (both fund types hold concentrated positions in stock i due to shared index exposure). The results below should be interpreted as consistent with the information channel story, not as a causal identification of it. We plan to implement a formal IV design with a reported first stage and instrument diagnostics in the revision.

Table 7 tests this prediction by including interactions of $OnLoan_{i,j,t-1}$ with $Disclosure_{i,t+1}$ (lead), $Disclosure_{i,t}$ (contemporaneous), and $Disclosure_{i,t-1}$ (lag). Column (3) shows that only the lead interaction is statistically significant (-0.8224, $t = -2.62$). The contemporaneous interaction is economically negligible (0.0353, $t = 0.08$) and the lagged interaction is small and insignificant (-0.1157, $t = -0.62$). Column (4) with saturated fixed effects confirms this pattern: the lead interaction is -0.8723 ($t = -2.27$), while contemporaneous and lagged interactions remain insignificant.

This timing pattern—pre-announcement selling by lenders but not in response to past or contemporaneous disclosure—is precisely what the information acquisition channel predicts. Active lending funds reduce positions in the month before a large short position becomes publicly known, consistent with observing borrowing demand before it crosses the public disclosure threshold.

Two interpretive notes are warranted. First, the ±3-month sample window means that $Disclosure_{i,t-1} = 1$ is observed in months when the large short position is already public information; in these months, any information advantage from lending has evaporated and position changes naturally revert toward zero, mechanically dampening the lagged interaction coefficient. The asymmetry test is therefore most credibly interpreted as showing that the lead interaction is significantly negative while the other interactions are near zero, rather than claiming the three coefficients are statistically different from each other. Formal Wald tests of $H_0\colon \beta_{\text{lead}} = \beta_{\text{lag}}$ and sensitivity analysis across alternative window widths are deferred to the revision. Second, the magnitude of the lagged interaction in column (4) is -0.3043 ($t = -1.34$), which is economically non-negligible and one-sided borderline significant; while it does not meet conventional significance thresholds, it is inconsistent with a clean zero and merits a formal test in the revision.

6 Information Channels within Fund Families

Having established that active funds trade on information acquired through securities lending, we now examine how this information flows within fund families. Fund families typically organize their lending operations through centralized lending desks, creating potential for information spillovers across funds managed by the same family or the same portfolio manager.

6.1 Decomposing Lending Sources

We decompose lending into three channels: (i) lending by funds managed by the same portfolio manager ($OnLoan_{i,m,j,t-1}$), (ii) lending by other funds within the same family but managed by a different manager ($OnLoan_{i,m,k\neq j,t-1}$), and (iii) lending by other funds in the family outside the manager's own fund group ($OnLoan_{i,family,k\neq m,t-1}$). This decomposition allows us to distinguish between information acquired directly by a manager through the manager's own funds and information that spills over from other parts of the fund family.

Table 8 presents the results. Columns (1)–(4) use security, fund, and time fixed effects; columns (5)–(6) use the saturated three-way fixed effects specification.

6.2 Same-Manager vs. Cross-Manager Lending

The results reveal a clear hierarchy in the strength of the information channel. The interaction of same-manager fund lending with the disclosure indicator is the strongest and most consistently significant, with coefficients ranging from -0.7283 to -1.9166 across specifications. This finding is intuitive: portfolio managers who directly observe borrowing demand through their own funds' lending positions have the most direct access to information.

Cross-manager lending within the same family (other funds in the family managed by different portfolio managers) shows a weaker but still detectable effect, with interaction coefficients around -0.1957 in the baseline specification. This suggests that information does spill

over across managers within fund families, likely through internal communication channels or centralized lending desk reports.

Family-level lending outside the manager's own funds shows intermediate effects (-0.5180 to -0.7800), confirming that information flows across the family organization but attenuates with organizational distance from the manager.

These results are consistent with the information flowing through the fund family's organizational structure, with the strongest signal coming from a manager's own funds and progressively weaker signals from more organizationally distant sources within the family. The hierarchy is intuitive in the framework of Nurisso [2026]: a portfolio manager who directly observes borrowing demand through their own funds' lending contracts has unmediated access to the private signal, while the same signal filtered through internal reporting channels or centralized desk summaries carries additional noise. The result that information attenuates with organizational distance within the family also connects to evidence that family-fund managers share investment ideas (Pool et al., 2012).

7 Market Structure and Information Acquisition

We next examine whether the informativeness of the lending signal varies with the structure of the lending market. Nurisso [2026] shows that in a Bertrand competition setting, a greater number of active lending agents reduces the informational content of each individual lending transaction: when short sellers can approach many lenders, the willingness of any single lender to engage reveals less about the size or conviction behind the short position. The corollary is that concentrated lending markets—with fewer active intermediaries—preserve a larger information share for each participating lender. If lending conveys private information about short-selling demand, the signal should therefore be more informative where lending is concentrated, amplifying the trading response documented in earlier sections.

We aggregate lending to the fund-family level for this analysis, using $OnLoan_{i,f,t-1}$—the total dollar value of stock i lent by all funds in family f at month $t-1$ divided by market capitalization, standardized to mean zero and unit variance. There are two reasons for this aggregation. First, market structure variation operates at the family level: the number of active lending agents, the HHI of lending concentration, and family portfolio breadth are inherently family-level constructs that capture the organizational unit through which lending decisions and information flow. Second, and more importantly, the concentration mechanisms we test—fewer counterparty relationships and more concentrated portfolios—generate more precise signals precisely because the family observes a larger share of total borrowing demand for a given stock. Fund-level lending would mechanically understate the information set available to a fund manager whose family coordinates lending through a centralized desk. In unreported robustness tests, the qualitative results hold when using fund-level lending instead, but are weaker in magnitude, consistent with family aggregation capturing the relevant information unit.

We test this by interacting family-level lending ($OnLoan_{i,f,t-1}$) with measures of lending market concentration and with the mandatory disclosure indicator.

Table 9 presents the results, all estimated with the full saturated fixed effects (security×time, fund×time, fund×security), active funds only, and standard errors double-clustered by fund and time. Column (1) reproduces the baseline: $OnLoan_{i,f,t-1} = -0.4226^{**}$ ($t = -2.35$) with a disclosure interaction of -0.8633^{**} ($t = -2.32$). The key variable throughout is *family-level* lending, $OnLoan_{i,f,t-1}$, measured as the aggregate dollar value of stock i lent by all funds in family f divided by market capitalization, standardized to mean zero and unit variance.

We consider five standardized measures of concentration. At the stock level: the Herfindahl–Hirschman index of lending concentration across families ($HHI^{stock}_{i,t-1}$, columns 2–3) and the inverse of the number of active lending agents ($1/ActiveAgents_{i,t-1}$, columns 4–5). At the family level: the family's own lending portfolio concentration ($HHI^{family}_{f,t-1}$, columns 6–7), the inverse of the number of stocks held ($1/StocksHeld_{f,t-1}$, columns 8–9), and the inverse of the

number of stocks lent ($1/StocksLent_{f,t-1}$, columns 10–11). For each concentration measure, we show a specification with only the pairwise interaction ($OnLoan \times Concentration$, even columns) and one adding the triple interaction with disclosure ($OnLoan \times Concentration \times Disclosure_{i,t+1}$, odd columns).

The results support the information concentration hypothesis. At the stock level, fewer active lending agents amplifies the signal: the triple interaction $OnLoan_{i,f,t-1} \times 1/ActiveAgents \times Disclosure_{i,t+1}$ is -9.2142^{**} ($t = -2.41$, column 5). When fewer agents intermediate the lending market for a stock, the private signal observed by each participating lender is correspondingly more informative about the specific identity and size of the short position being built.

Family-level concentration further amplifies the effect. The triple interaction with family HHI is -3.0048^{*} ($t = -1.94$, column 7): families that concentrate their lending in fewer stocks hold more concentrated positions in each borrower relationship, making each lending contract more informative. The strongest result comes from the inverse of stocks lent: $OnLoan \times 1/StocksLent \times Disclosure_{i,t+1} = -8.1425^{***}$ ($t = -3.96$, column 11). When a family lends fewer distinct securities, each lending relationship provides a larger information share about short selling activity in that stock.

These heterogeneity results reinforce the information acquisition interpretation. The trading response to lending signals is amplified precisely where information concentration is highest—a pattern that is difficult to reconcile with mechanical alternatives and is instead consistent with fund managers extracting and acting on informative signals embedded in the borrowing demand observed through their lending operations. The findings also have implications for market design: policy interventions that fragment the lending market (reducing concentration) may diminish the informational efficiency gains that lending currently provides to active fund managers.

8 Conclusion

We document that the securities lending market serves as an information acquisition channel for active mutual fund managers. Using unique regulatory data from the Deutsche Bundesbank with security-level lending positions, we show that active fund managers reduce their holdings in stocks they lend. The effect is economically large—a 0.33 percentage point position reduction per one-standard-deviation increase in lending activity—and is absent for passive funds, who cannot adjust their portfolios in response to the signal.

We identify the mechanism through mandatory short position disclosure events under EU regulation. Active funds reduce positions substantially more when lending precedes a disclosure event, confirming that the information content of lending drives the trading response. The timing is consistent with anticipation: only the lead disclosure amplifies the effect, while contemporaneous and lagged disclosure interactions are small and statistically insignificant. Results are robust to using passive fund lending within the same fund family as an alternative lending measure.

We further show that information flows within fund families, with the strongest effects from same-manager fund lending. The trading response is amplified in concentrated lending markets where individual transactions carry more informational weight.

Our findings contribute to the understanding of how information flows through financial markets. The securities lending market, typically studied from the perspective of short sellers, also functions as an information channel for the long side of the market. Fund managers who participate in lending gain private, real-time information about short-selling demand that enables them to make more informed portfolio decisions.

These results have implications for the welfare distribution in financial markets. Active fund managers who lend benefit from an informational advantage that allows them to reduce positions before adverse price movements materialize, generating gains for their investors at the expense of uninformed buyers on the other side of the trade. Short sellers, in turn, may face higher borrowing costs or reduced lending supply if lenders anticipate acting on the information,

consistent with the commitment problem modeled by Nurisso [2026] and the evidence in Palia and Sokolinski [2024] that short sellers prefer passive lenders precisely because they cannot exploit the signal. Market concentration amplifies these information asymmetries: in thin lending markets, the information advantage accrues to a small number of participating families, with distributional consequences for price efficiency.

Our study examines the period December 2014–December 2018, prior to the implementation of the Securities Financing Transactions Regulation (SFTR), which took effect in 2020 and introduced transaction-level reporting requirements for securities lending. SFTR materially increases the transparency of the lending market and may attenuate the private information advantage we document. Whether the lending-information channel survives under SFTR's enhanced disclosure regime is an important question for future research.

These results have broader implications for the regulation of securities lending, the organization of fund families, and the design of mandatory disclosure regimes. Our evidence suggests that lending market structure—particularly the degree of concentration—is a first-order determinant of how much private information flows to active managers, with implications for policymakers considering reforms to lending market infrastructure.

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Table 1
Descriptive Statistics: German Mutual Funds - Investment Focus
This table reports descriptive statistics of the sample of German mutual funds used in the analysis. Data on fund portfolio holdings are from the Bundesbank's Investment Funds Statistics database. Other fund information is from Morningstar.

Investment Focus	*Number of Funds*	*AUM (in billion EUR)*
Global	92	63.78
Europe	135	38.37
Germany	58	49.40
Other	35	5.20
ALL	**320**	**156.75**

Table 2
Descriptive Statistics: German Mutual Funds - Securities Lending Behavior
This table reports descriptive statistics of the sample of German mutual funds used in the analysis. Data on fund portfolio holdings are from the Bundesbank's Investment Funds Statistics database. Other fund information is from Morningstar.

Panel A: Fund-Level Lending	Obs	% of Funds Lending
All Funds	320	32.81%
Active Funds	267	25.47%
Passive Funds	53	69.81%

Panel B: Portfolio-Level and Position-Level Lending	Obs	Mean	Median	St. Dev.	25th Pct.	75th Pct.
% of AUM Lent	2,563	3.80%	2.43%	4.50%	0.94%	5.12%
% of Position MV Lent	21,572	55.39%	57.44%	36.20%	17.64%	92.77%

Table 3
Descriptive Statistics
This table reports monthly descriptive statistics of the main variables used in the analysis. Data on fund portfolio holdings and equity lending activities is from the Bundesbank's Investment Funds Statistics database. Stock data is from Datastream. Markit provides equity market lending data. Fund information is from Morningstar. The variable definitions are in Appendix A.

Variable	Mean	Median	St. Dev.	25th Pct.	75th Pct.	Obs.
Panel A: Portfolio Variables						
Position Change (in %)	0.94	0.00	16.15	0.00	0.00	1,022,668
Position Weight (in %)	1.08	0.63	1.35	0.24	1.42	1,022,668
$D(Lender)_{i,j,t}$	0.02	-	0.14	-	-	1,022,668
$OnLoan_{i,j,t}$ *(% of Market Cap.)*	6.86	1.36	26.18	0.15	5.34	19,324
$D(Lender)_{i,f,t}$	0.05	-	0.22	-	-	1,022,668
$OnLoan_{i,f,t}$ *(% of Market Cap.)*	7.65	1.32	29.65	0.11	6.50	50,729
Panel B: Stock Variables						
On Loan (% of Market Cap.)	2.46	1.05	3.83	0.38	2.80	134,909
Lending Supply (% of Market Cap.)	18.18	17.23	10.14	9.98	25.88	134,865
Short Fee (bps p.a.)	65.54	18.81	217.29	11.32	27.46	131,390
Return (in %)	0.65	0.54	8.94	-4.16	5.38	137,321
Carhart four-factor Alpha (in %)	0.11	0.04	8.60	-4.12	4.22	129,857
D(Disclosed Short Position)	0.19	-	0.39	-	-	137,374
D(Disclosure Event)	0.00	-	0.06	-	-	137,374
Panel C: Fund Variables						
D(Active Fund)	0.85	-	0.36	-	-	13,010
D(Fund Lends)	0.20	-	0.40	-	-	13,010
Gross Return (in %)	0.41	0.66	3.99	-1.91	3.04	12,876
Net Return (in %)	0.29	0.56	3.98	-2.02	2.91	12,876
Gross Alpha (Carhart four-factor) (in %)	-0.02	0.00	1.60	-0.88	0.84	12,840
Net Alpha (Carhart four-factor) (in %)	-0.13	-0.11	1.60	-0.98	0.73	12,861
Fund R^2 *(in %)*	86.52	89.00	10.49	82.77	93.89	12,849
Fund Costs (excl. Transaction Fees) (in %)	1.36	1.46	0.69	0.96	1.72	12,879
Net Asset Value (in billion €)	0.56	0.10	1.59	0.03	0.34	13,010
Fund Flows (in % of AUM)	0.59	-0.05	28.86	-0.62	0.50	13,010
Fund Age (in years)	18.60	16.25	13.08	9.75	21.58	13,010
Fund Turnover (in %)	4.48	1.95	8.21	0.08	5.44	13,010
Panel D: Fund Family Variables						
D(Family Lends)	0.17	-	0.37	-	-	1,543
Family Net Asset Value (in billion €)	45.79	12.40	72.00	3.47	49.87	1,543

Table 4
Position Changes and Securities Lending
This table reports results from panel regressions of monthly position changes ($PosChange_{i,j,t}$) on lagged securities lending indicators. Columns (1)–(2) use the full sample; columns (3)–(5) restrict to active funds. All specifications include security×time, fund×time, and fund×security fixed effects. Standard errors are clustered at the fund level. t-statistics are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels. Variable definitions and detailed notes appear below the table.

	All Funds		Active Funds		
	(1)	(2)	(3)	(4)	(5)
$OnLoan_{i,j,t-1}$	-0.2069^{***}	0.0133	-0.3271^{***}		-0.3206^{***}
	(−4.23)	(0.80)	(−5.39)		(−5.49)
$Active_j \times OnLoan_{i,j,t-1}$		-0.3382^{***}			
		(−5.33)			
$D(Lender)_{i,j,t-1}$				-1.9165^{***}	−0.0584
				(−4.18)	(−0.15)
Security × Time FE	Yes	Yes	Yes	Yes	Yes
Fund × Time FE	Yes	Yes	Yes	Yes	Yes
Fund × Security FE	Yes	Yes	Yes	Yes	Yes
SE Clustering	Fund	Fund	Fund	Fund	Fund
Sample	All	All	Active	Active	Active
Adj. R^2	0.2483	0.2483	0.1713	0.1713	0.1713
N	1,022,668	1,022,668	757,092	757,092	757,092

Notes: The table reports results from panel regressions of $PosChange_{i,j,t}$, the monthly percentage change in the number of shares of stock i held by fund j at time t. $OnLoan_{i,j,t-1}$ is the dollar value of shares lent by fund j in stock i at month $t-1$ divided by the market capitalization of stock i, standardized to mean zero and unit standard deviation. $Active_j$ equals one for actively managed funds. $D(Lender)_{i,j,t-1}$ equals one if fund j lent any shares of stock i at $t-1$. Controls include the lagged Amihud (2002) illiquidity percentile rank and the lagged position weight within the fund's portfolio. Observations where fund j is entering position in stock i for the first time are excluded. All specifications include security×time, fund×time, and fund×security fixed effects, absorbing stock-level price signals, fund-level portfolio decisions, and fund-stock fixed preferences, respectively. Standard errors are clustered at the fund level. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 5
Securities Lending and Mandatory Short Position Disclosure

This table reports panel regressions of $PosChange_{i,j,t}$, the monthly percentage change in fund j's position in stock i, estimated on the *disclosure sample* — observations within ±3 months of a first-time public short position disclosure for stocks ever receiving such a disclosure ($ever_LSP = 1$). $OnLoan_{i,j,t-1}$ is the dollar value of shares lent by fund j in stock i at month $t-1$ scaled by stock i's market capitalization, standardized to mean zero and unit standard deviation. $Disclosure_{i,t+1}$ equals one if stock i receives a *new* mandatory public short position disclosure in month $t+1$ (no prior disclosure in the preceding six months). Columns (1)–(7) estimate on actively managed funds; column (8) is the passive fund placebo. Columns (1)–(3) and (5)–(7) include separate security, fund, and time fixed effects; columns (4) and (8) use the fully saturated specification (security×time, fund×time, and fund×security). Controls include lagged Amihud illiquidity rank and lagged position weight.

	Active Funds							Passive Funds
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$OnLoan_{i,j,t-1}$	-0.2837^{***}	-0.2835^{***}	-0.1868	-0.2321	-0.2848^{***}	-0.2841^{***}	-0.2124	0.0208
	(−2.85)	(−2.86)	(−1.63)	(−0.87)	(−2.87)	(−2.85)	(−1.53)	(0.23)
$Disclosure_{i,t+1}$		0.3633	0.3398			0.3618	0.3391	
		(1.25)	(1.17)			(1.25)	(1.17)	
$OnLoan_{i,j,t-1} \times Disclosure_{i,t+1}$			-0.8027^{**}	-0.8622^{**}			-0.8031^{**}	0.0258
			(−2.68)	(−2.22)			(−2.68)	(0.28)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Security FE	Yes	Yes	Yes		Yes	Yes	Yes	
Fund FE	Yes	Yes	Yes		Yes	Yes	Yes	
Time FE	Yes	Yes	Yes		Yes	Yes	Yes	
Security × Time FE				Yes				Yes
Fund × Time FE				Yes				Yes
Fund × Security FE				Yes				Yes
SE Clustering	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time
Sample	Active	Active	Active	Active	Active	Active	Active	Passive
Adj. R^2	0.3288	0.3289	0.3291	0.8585	0.3289	0.3290	0.3292	0.8585
N	42,512	42,512	42,512	39,109	42,512	42,512	42,512	11,288

Notes: Standard errors are double-clustered by fund and time. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 6
Passive Fund Lending as Reduced-Form Robustness Check
This table reports reduced-form results replacing fund-level lending with passive fund lending ($OnLoan_{i,f\text{-}passive,t-1}$) as an alternative lending measure. Since passive funds track indices and cannot trade on information, their lending reflects portfolio composition and borrower demand rather than active rebalancing. This is a robustness exercise; formal IV estimation with a reported first stage is deferred to the revision. The dependent variable is position changes for active funds. Standard errors are double-clustered by fund and time. t-statistics are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels. Variable definitions and detailed notes appear below the table.

	(1)	(2)	(3)	(4)
$OnLoan_{i,f\text{-passive},t-1}$	-0.3406^{***}	-0.3399^{***}	-0.2005^{*}	-0.4475^{**}
	(-3.20)	(-3.18)	(-1.78)	(-2.35)
$Disclosure_{i,t+1}$		0.3600	0.3304	
		(1.25)	(1.16)	
$OnLoan_{i,f\text{-passive},t-1} \times Disclosure_{i,t+1}$			-1.1240^{***}	-0.9156^{**}
			(-3.29)	(-2.32)
Controls	Yes	Yes	Yes	Yes
Security FE	Yes	Yes	Yes	
Fund FE	Yes	Yes	Yes	
Time FE	Yes	Yes	Yes	
Security × Time FE				Yes
Fund × Time FE				Yes
Fund × Security FE				Yes
SE Clustering	Fund, Time	Fund, Time	Fund, Time	Fund, Time
Sample	Active	Active	Active	Active
N	42,512	42,512	42,512	39,109

Notes: The table reports results from panel regressions of $PosChange_{i,j,t}$, the monthly percentage change in fund j's position in stock i, using passive fund lending as an exogenous measure of short selling demand. The sample is restricted to actively managed funds in the disclosure sample. $OnLoan_{i,f\text{-passive},t-1}$ is the aggregate dollar value of stock i lent by *passive* funds in family f at month $t-1$, divided by the market capitalization of stock i, standardized to mean zero and unit standard deviation. Because passive funds track benchmark indices and cannot trade on information, their lending activity is driven by portfolio composition and borrower demand rather than strategic portfolio decisions, providing an exogenous source of variation in short selling demand signals available to the family. $Disclosure_{i,t+1}$ equals one if stock i has a new mandatory public short position disclosure in month $t+1$ (no prior disclosure in the preceding six months). Controls include lagged Amihud illiquidity rank and lagged position weight. Standard errors are double-clustered by fund and time. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 7
Leads and Lags of Mandatory Disclosure
This table reports results including leads, contemporaneous, and lagged disclosure indicators and their interactions with lending. The dependent variable is position changes for active funds. Standard errors are double-clustered by fund and time. t-statistics are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels. Variable definitions and detailed notes appear below the table.

	(1)	(2)	(3)	(4)
$OnLoan_{i,j,t-1}$	-0.2837^{***}	-0.2820^{***}	-0.1671	-0.2110
	(-2.85)	(-2.83)	(-1.31)	(-0.67)
$Disclosure_{i,t+1}$		0.4168	0.3928	
		(1.40)	(1.31)	
$Disclosure_{i,t}$		0.2958	0.3013	
		(1.05)	(1.07)	
$Disclosure_{i,t-1}$		-0.0972	-0.1020	
		(-0.29)	(-0.30)	
$OnLoan_{i,j,t-1} \times Disclosure_{i,t+1}$			-0.8224^{**}	-0.8723^{**}
			(-2.62)	(-2.27)
$OnLoan_{i,j,t-1} \times Disclosure_{i,t}$			0.0353	0.3738
			(0.08)	(0.55)
$OnLoan_{i,j,t-1} \times Disclosure_{i,t-1}$			-0.1157	-0.3043
			(-0.62)	(-1.34)
Controls	Yes	Yes	Yes	Yes
Security FE	Yes	Yes	Yes	
Fund FE	Yes	Yes	Yes	
Time FE	Yes	Yes	Yes	
Security × Time FE				Yes
Fund × Time FE				Yes
Fund × Security FE				Yes
SE Clustering	Fund, Time	Fund, Time	Fund, Time	Fund, Time
Sample	Active	Active	Active	Active
N	42,512	42,522	42,512	39,109

Notes: Panel regressions of $PosChange_{i,j,t}$ on leads, contemporaneous, and lagged mandatory disclosure indicators interacted with lending. Sample: active funds in the disclosure subsample. $OnLoan_{i,j,t-1}$: dollar value lent by fund j in stock i at $t-1$ divided by market cap, standardized. $Disclosure_{i,t+1}$: indicator for a *new* mandatory public short-position disclosure for stock i in month $t+1$ (no prior disclosure in the preceding six months); $Disclosure_{i,t}$: contemporaneous; $Disclosure_{i,t-1}$: lagged. Under the information acquisition hypothesis, only $OnLoan \times Disclosure_{i,t+1}$ should be significant. Controls: lagged Amihud illiquidity rank and lagged position weight. Column (2) has $N = 42{,}522$ versus 42,512 in columns (3)–(4) because ten observations have defined disclosure indicators but missing $OnLoan$; these are excluded once interaction terms are added. SE double-clustered by fund and time. t-statistics in parentheses. *, **, *** denote 10%, 5%, 1% significance.

Table 8
Information Channels within Fund Families

This table reports panel regressions of $PosChange_{i,j,t}$ on lending-based information channels decomposed within fund families, using the disclosure sample of actively managed funds. Panel A indicates which channel variables are included; Panel B reports their interactions with $Disclosure_{i,t+1}$. Lending is decomposed into four channels: $OnLoan_{i,t-1}$ (market-wide aggregate); $OnLoan_{i,m,j,t-1}$ (stock i lent by other funds sharing fund j's portfolio manager); $OnLoan_{i,m,k\neq j,t-1}$ (stock i lent by family funds under a different manager); $OnLoan_{i,f,k\neq m,t-1}$ (stock i lent by family funds outside the manager's group entirely). $Disclosure_{i,t+1}$ equals one if stock i receives a new mandatory public short position disclosure in month $t+1$. All lending variables are standardized. Columns (1)–(4) include separate security, fund, and time fixed effects; columns (5)–(6) use the fully saturated specification (security×time, fund×time, and fund×security). Controls include lagged Amihud illiquidity rank and lagged position weight.

	Security + Fund + Time FE				Saturated FE	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Lending channel variables included						
$OnLoan_{i,t-1}$ (market aggregate)	Yes	Yes	Yes	Yes	Yes	Yes
$OnLoan_{i,m,j,t-1}$ (same-manager funds)		Yes	Yes	Yes	Yes	Yes
$OnLoan_{i,m,k\neq j,t-1}$ (other funds, same manager)			Yes	Yes	Yes	Yes
$OnLoan_{i,f,k\neq m,t-1}$ (family, other managers)				Yes		Yes
Panel B: Interactions with $Disclosure_{i,t+1}$						
Same-manager × $Disclosure_{i,t+1}$		−0.7283*	−0.8015*	−0.9245**	−1.5630***	−1.9166***
		(−1.72)	(−1.89)	(−2.15)	(−2.90)	(−3.44)
Other-fund × $Disclosure_{i,t+1}$			−0.1957**	−0.1957**	−0.1204	−0.1133
			(−2.04)	(−2.04)	(−1.08)	(−1.01)
Family outside manager × $Disclosure_{i,t+1}$				−0.5180*		−0.7800**
				(−1.75)		(−2.18)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Security × Time FE					Yes	Yes
Fund × Time FE					Yes	Yes
Fund × Security FE					Yes	Yes
SE Clustering	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time	Fund, Time
N	41,145	41,145	41,145	41,145	39,675	39,675

Notes: Standard errors are double-clustered by fund and time. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

A Variable Definitions

B Additional Tables

Table 9
Market Structure and Information Acquisition

This table reports panel regressions of $PosChange_{i,j,t}$ on $OnLoan_{i,f,t-1}$ (aggregate family lending of stock i scaled by market capitalization, standardized) and its interactions with lending market concentration measures, using the disclosure sample of actively managed funds. $Disclosure_{i,t+1}$ equals one if stock i receives a new mandatory public disclosure in month $t+1$. Panel A reports the baseline main effects. Panel B interacts with stock-level concentration: $HHI^{stock}_{i,t-1}$ is the Herfindahl–Hirschman index of lending concentration across families for stock i; $1/ActiveAgents_{i,t-1}$ is the inverse number of active lending agents. Panel C interacts with family-level concentration: $HHI^{family}_{f,t-1}$ is the HHI of family f's lending portfolio across stocks; $1/StocksHeld_{f,t-1}$ is the inverse number of stocks held by family f ($\times 10^4$); $1/StocksLent_{f,t-1}$ is the inverse number of stocks lent by family f ($\times 10^4$). All concentration measures are standardized. All specifications include security×time, fund×time, and fund×security fixed effects. Controls include lagged Amihud illiquidity rank and lagged position weight.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: Main effects and baseline interactions											
$OnLoan_{i,f,t-1}$	−0.4226**										
	(−2.35)										
$OnLoan_{i,f,t-1} \times Disclosure_{i,t+1}$	−0.8633**										
	(−2.32)										
Panel B: Interactions with stock-level concentration											
$OnLoan_{i,f,t-1} \times HHI^{stock}_{i,t-1}$		−0.2148									
		(−1.18)									
$OnLoan_{i,f,t-1} \times HHI^{stock}_{i,t-1} \times Disclosure_{i,t+1}$			−1.3854*								
			(−1.80)								
$OnLoan_{i,f,t-1} \times 1/ActiveAgents_{i,t-1}$				−1.5832*							
				(−1.75)							
$OnLoan_{i,f,t-1} \times 1/ActiveAgents_{i,t-1} \times Disclosure_{i,t+1}$					−9.2142**						
					(−2.41)						
Panel C: Interactions with family-level concentration											
$OnLoan_{i,f,t-1} \times HHI^{family}_{f,t-1}$						−0.1973					
						(−1.42)					
$OnLoan_{i,f,t-1} \times HHI^{family}_{f,t-1} \times Disclosure_{i,t+1}$							−3.0048*				
							(−1.94)				
$OnLoan_{i,f,t-1} \times 1/StocksHeld_{f,t-1}$								−0.5812*			
								(−1.68)			
$OnLoan_{i,f,t-1} \times 1/StocksHeld_{f,t-1} \times Disclosure_{i,t+1}$									−10.4297*		
									(−1.72)		
$OnLoan_{i,f,t-1} \times 1/StocksLent_{f,t-1}$										−1.0463**	
										(−2.31)	
$OnLoan_{i,f,t-1} \times 1/StocksLent_{f,t-1} \times Disclosure_{i,t+1}$											−8.1425***
											(−3.96)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Security × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Security FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SE Clustering						Fund, Time					
Sample						Active					
N	39,109	39,003	39,003	39,003	39,003	39,003	39,003	39,003	39,003	39,003	39,003

Note: Col. (1): N = 39,109; cols. (2)–(11): N = 39,003; 106 observations lack defined concentration measures.

Notes: Standard errors are double-clustered by fund and time. t-statistics in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 10
Variable Definitions

Variable	Definition
Panel A: Portfolio Variables	
$PosChange_{i,j,t}$	Percentage change in the number of shares of stock i held by fund j in month t, adjusted for stock splits. Winsorized at the 99.5% level.
$Position\ Weight_{i,j,t}$	Market value of holding i by fund j at time t divided by total fund assets under management.
$D(Position\ Exit)_{i,j,t}$	Indicator equal to one if fund j closes its position in stock i in month t.
$D(On\ Loan)_{i,j,t}$	Indicator equal to one if fund j has lent (part of) its position in stock i in month t.
$sh_sec_lend_{i,j,t}$	Fraction of the position in stock i lent by fund j in month t.
$OnLoan_{i,j,t}$	Dollar value of stock i lent by fund j in month t divided by stock i's total market capitalization. This is the fund-security-level lending variable used in all regression specifications; standardized to mean zero and unit standard deviation in estimation. Related to $sh_sec_lend_{i,j,t}$ by the fund's portfolio weight in stock i.
Panel B: Stock Variables	
$OnLoan_{i,t}$	Value of stock i on loan divided by total market capitalization. Source: IHS Markit.
$Lending\ Supply_{i,t}$	Value of stock i available for lending divided by total market capitalization. Source: IHS Markit.
$Short\ Fee_{i,t}$	Annualized fee (in basis points) for borrowing stock i. Source: IHS Markit.
$Disclosure_{i,t}$	Indicator equal to one if a mandatory public disclosure of a net short position exceeding 0.5% of issued share capital is filed for stock i at time t under Regulation (EU) No 236/2012. In regressions, $Disclosure_{i,t+1}$ equals one only for *new* disclosures (no prior disclosure in the preceding six months).
Panel C: Fund Variables	
$D(Active\ Fund)_j$	Indicator equal to one if fund j is actively managed.
$D(Fund\ Lends)_{j,t}$	Indicator equal to one if fund j has at least one position on loan in month t.
$Net\ Asset\ Value_{j,t}$	Total assets under management of fund j in month t, aggregated across share classes.
$Fund\ Flows_{j,t}$	Net fund flows as a percentage of beginning-of-month AUM.
Panel D: Fund Family Variables	
$D(Family\ Lends)_{f,t}$	Indicator equal to one if at least one fund in family f has a position on loan in month t.
$Family\ NAV_{f,t}$	Total assets under management across all funds in family f in month t.